

Lab 3 — Azure Key Vault Security & RBAC

SC-500 Azure Security Engineering Portfolio Lab

Resource Group	rg-security-lab
Key Vault	kv-security-lab
Region	East US
Permission Model	Azure RBAC
RBAC Role	Key Vault Secrets Officer
Lab User	labuser01
Test Secret	Demo-Lab-Secret2026

Objective

Demonstrate secure Azure Key Vault deployment, Azure RBAC-based access control, secret management, and least-privilege access.

Security Configuration

- Soft-delete enabled with a 90-day retention period.
- Purge protection enabled.
- Azure role-based access control (RBAC) selected as the permission model.
- Standard Key Vault tier deployed in East US.
- Public endpoint retained for the controlled lab environment.

RBAC Configuration

The lab identity **labuser01** was assigned the **Key Vault Secrets Officer** role at the **kv-security-lab** scope. A Key Vault Reader assignment was also used to support portal visibility of Key Vault objects. This demonstrates least-privilege access rather than broad subscription-level administrative permissions.

Secret Management & Validation

Using labuser01, the test secret **Demo-Lab-Secret2026** was successfully created and enabled. The secret was then opened successfully, confirming that the configured RBAC permissions allowed secret access. The secret value is intentionally not included in this portfolio document.

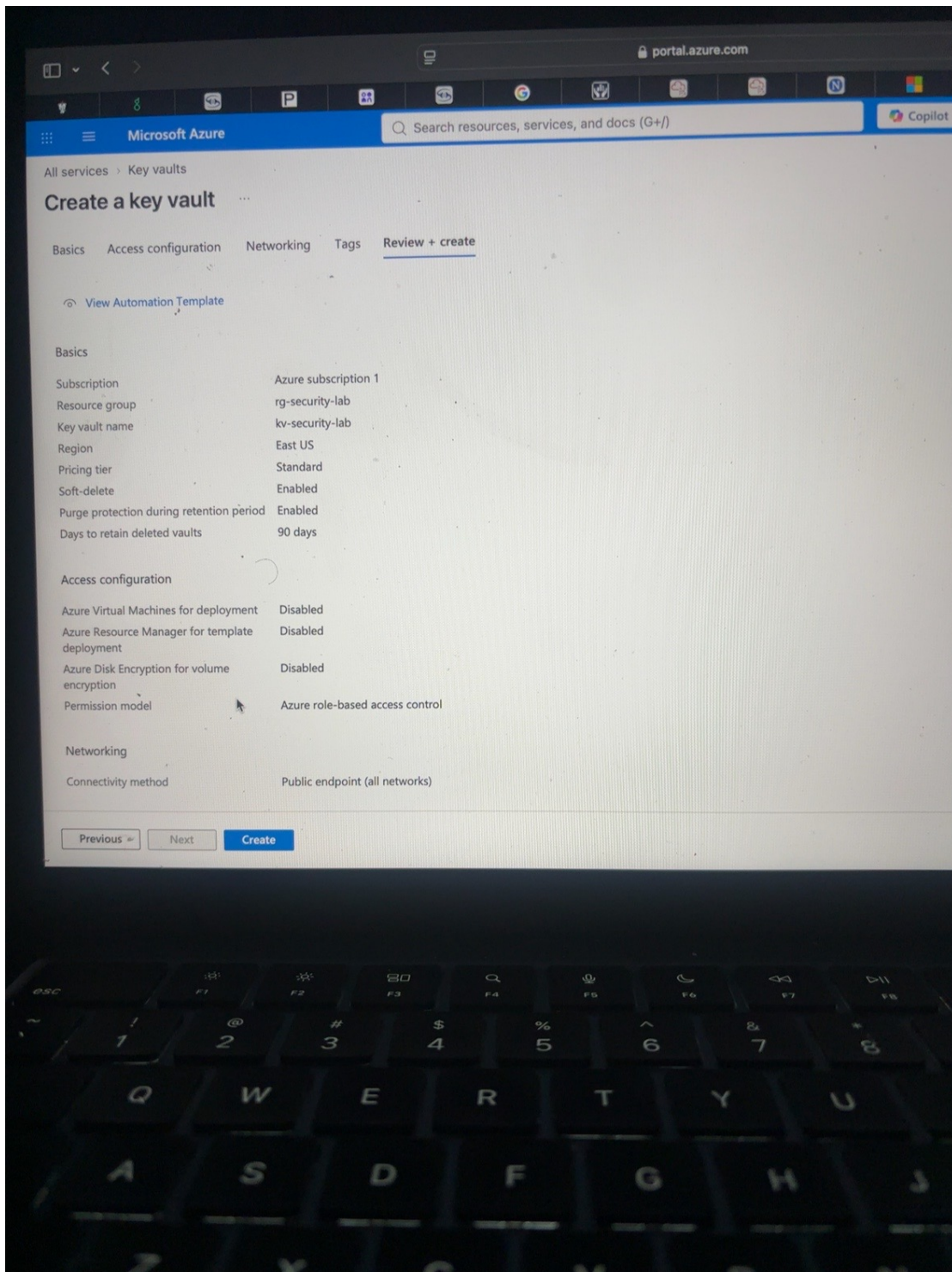
Security Concepts Demonstrated

- Azure Key Vault and cloud secrets management
- Azure RBAC and least privilege
- Microsoft Entra ID identity-based access control
- Soft-delete and purge protection

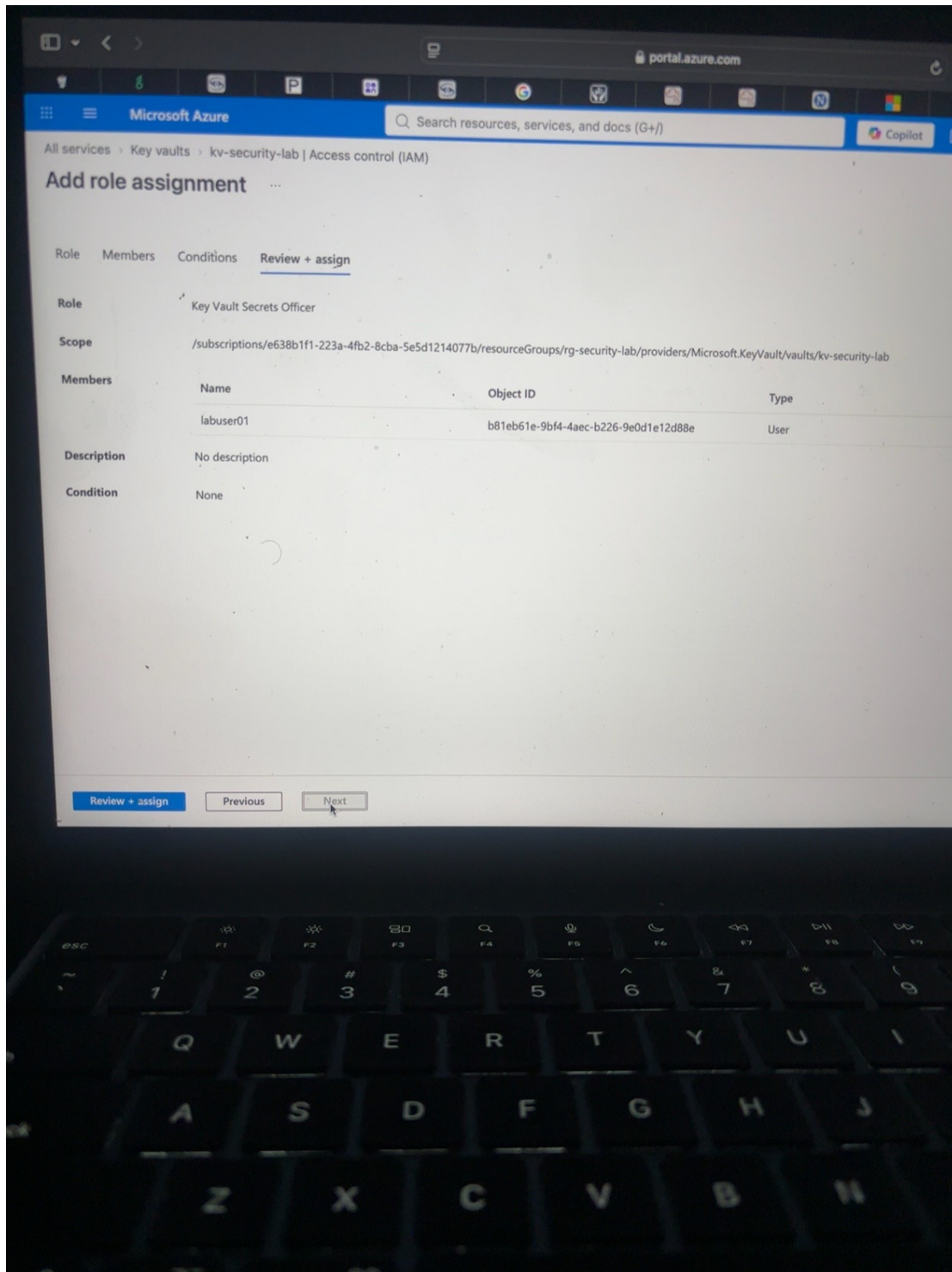
- Security validation and evidence collection
- Safe handling of sensitive information

Evidence

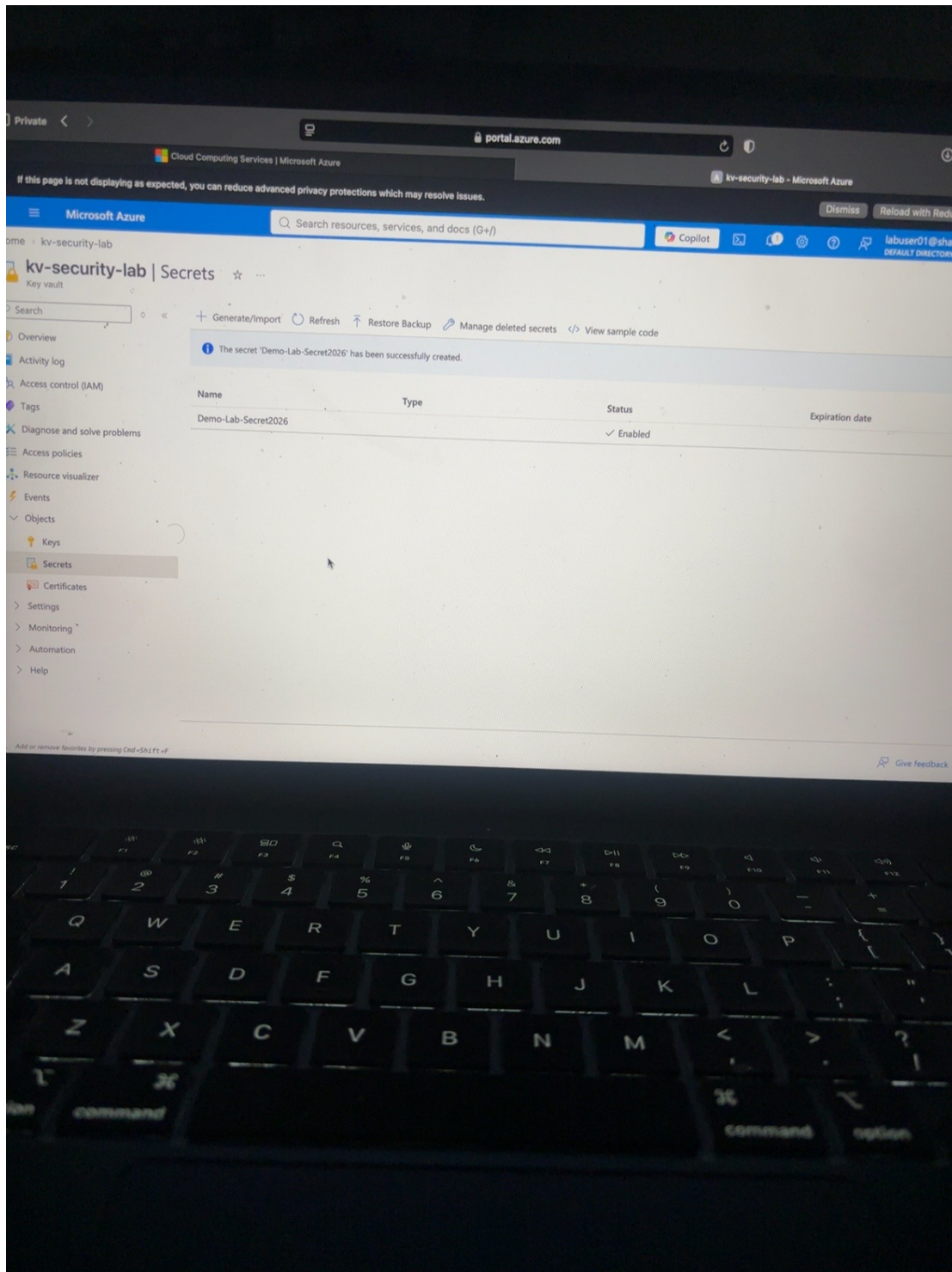
Evidence 1 — Key Vault security configuration



Evidence 2 — Key Vault Secrets Officer RBAC assignment



Evidence 3 — Secret successfully created and enabled



Note: The screenshot used for portfolio evidence shows the secret name and enabled status. The separate screenshot where the secret value was visible is intentionally excluded from this PDF.